SIR MV Young Innovators Challenge

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Applicant Type: Individual innovator

Patent Application Number: 202521094612

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Title of Invention: Fibélle Banana Fiber Yoga Mat

Video link: https://youtu.be/vo5iDqs1oMU **School:** Apeejay School, Kharghar, Navi Mumbai

Innovation category: Environment
Sub-category: Waste Management and Recycling

Fibélle Banana Fiber Yoga Mat Empowering Green Living (Patent Application Number: 202521094612):

Synthetic yoga mats are made of PVC (polyvinyl chloride), which releases harmful chemicals during production and disposal. They release Strong chemical smell when new. During work out slip occurs due excessive sweating. Cotton yoga mat Cotton absorbs sweat but becomes slippery when wet. They are generally thin, so not enough cushioning for joints. To overcome these problems this innovation(Patent Application Number: 202521094612) proposes yoga mat made up of majorly banana fiber. This newly developed mat solves slip problem of mats and eco-hazard of PVC mats. Banana fibers are naturally absorbent but dry faster than cotton. Therefore, it helps to keep sweat away while still maintaining grip. This yoga mat offers grip, durability and cushioning. Also, this mat aligns with global eco-conscious consumer trends. This innovation aligns with 'Sustainability Development Goal (**SDG) 3**' i.e. 'Good health and well-being' Fibélle yoga mat introduces a multi-layered yoga mat engineered with: Banana fiber fabric, finished with a vegan suede-like plant coating for comfort and grip, A banana foam–latex cushioning layer for enhanced resilience and joint support, and A cork–banana composite base with reinforced joints to provide slip resistance and long-term stability. Fibélle yoga mat introduces a multi-layered yoga mat engineered with: Banana fiber fabric, finished with a vegan suede-like plant coating for comfort and grip, A banana foam–latex cushioning layer for enhanced resilience and joint support, and banana composite base with reinforced joints to provide slip resistance and long-term stability.



Fibélle Banana Fiber Yoga Mat (Patent Application Number: 202521094612)
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Problem and solution

Problem

Why Replace Conventional Yoga Mats?

- PVC mats contribute to plastic pollution and carbon emissions
- Chemical smell affects user comfort
- Slippery surface during sweaty sessions may lead to falls
- Insufficient cushioning can strain joints during practice

Solution

Fibélle Banana Fiber Yoga Mat
(Patent Application No. 202521094612)

- Crafted from renewable banana fiber and plant-based materials
- Naturally absorbs sweat while maintaining a firm grip
- Multi-layer cushioning for lasting comfort and joint protection
- Eco-conscious, durable, and designed for healthier living



https://youtu.be/vo5iDqs1oMU

Fibélle Banana Fiber Yoga Mat

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Layer 1: Core Cushioning
Layer 2: Top Surface
Layer 3: Base Layer

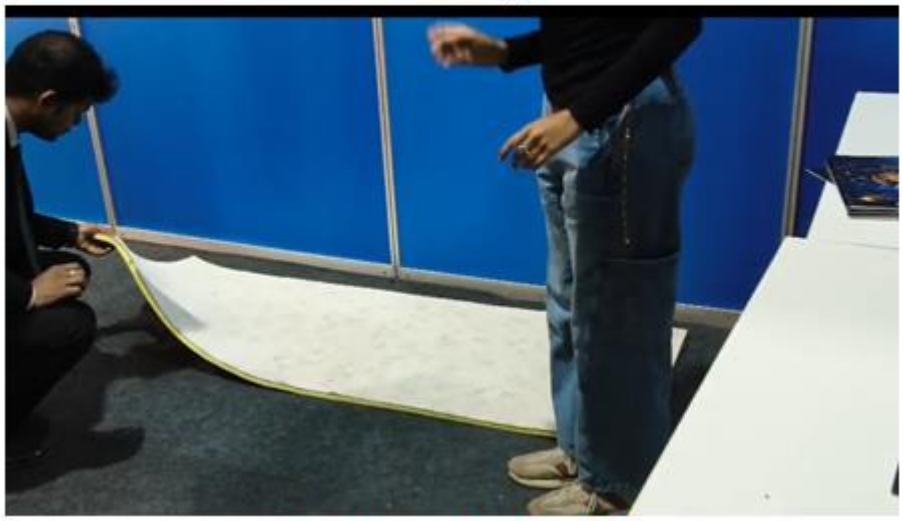


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Product Images



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Product Images



Fibélle Banana Fiber Yoga Mat